

MATH 4820 / Regression Analysis

Instructor: Mehdi Maadooliat

CHAPTER 1

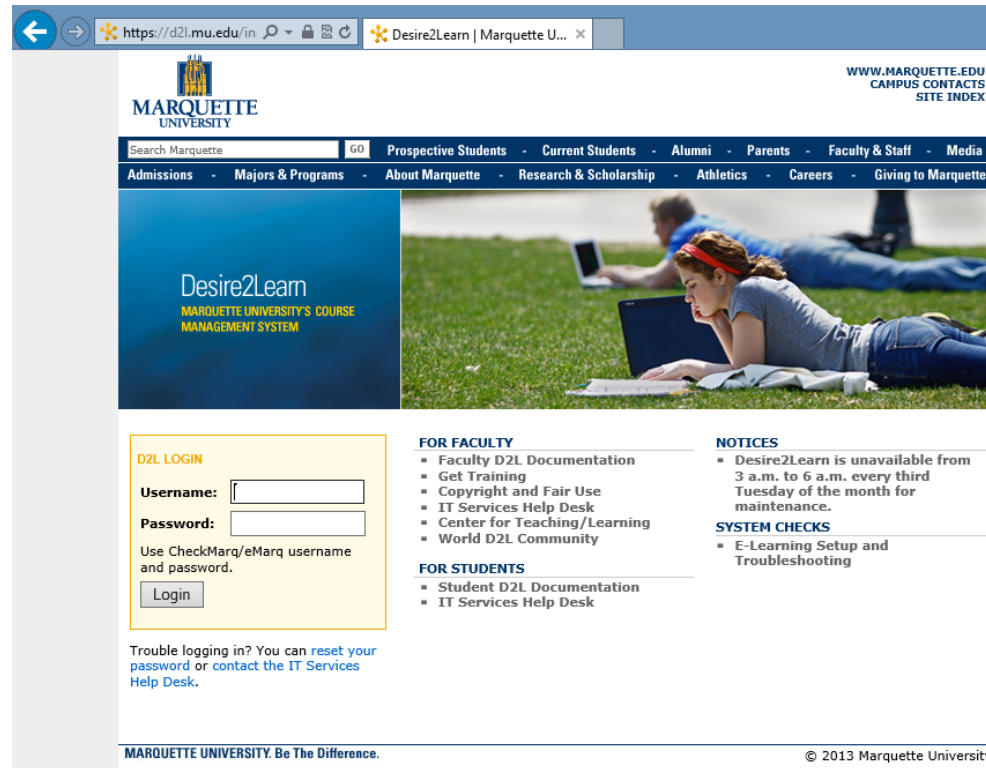
Introduction



Department of Mathematics, Statistics and Computer Science

SYLLABUS - D2L - ONLINE MATERIALS

- Syllabus
- Go to <https://d2l.mu.edu/index.asp>
- Enter your account info and click on **Log in!**



Marquette University logo and navigation menu.

Desire2Learn
MARQUETTE UNIVERSITY'S COURSE
MANAGEMENT SYSTEM

D2L LOGIN

Username:

Password:

Use CheckMarq/eMarq username and password.

Login

Trouble logging in? You can [reset your password](#) or [contact the IT Services Help Desk](#).

FOR FACULTY

- Faculty D2L Documentation
- Get Training
- Copyright and Fair Use
- IT Services Help Desk
- Center for Teaching/Learning
- World D2L Community

FOR STUDENTS

- Student D2L Documentation
- IT Services Help Desk

NOTICES

- Desire2Learn is unavailable from 3 a.m. to 6 a.m. every third Tuesday of the month for maintenance.

SYSTEM CHECKS

- E-Learning Setup and Troubleshooting

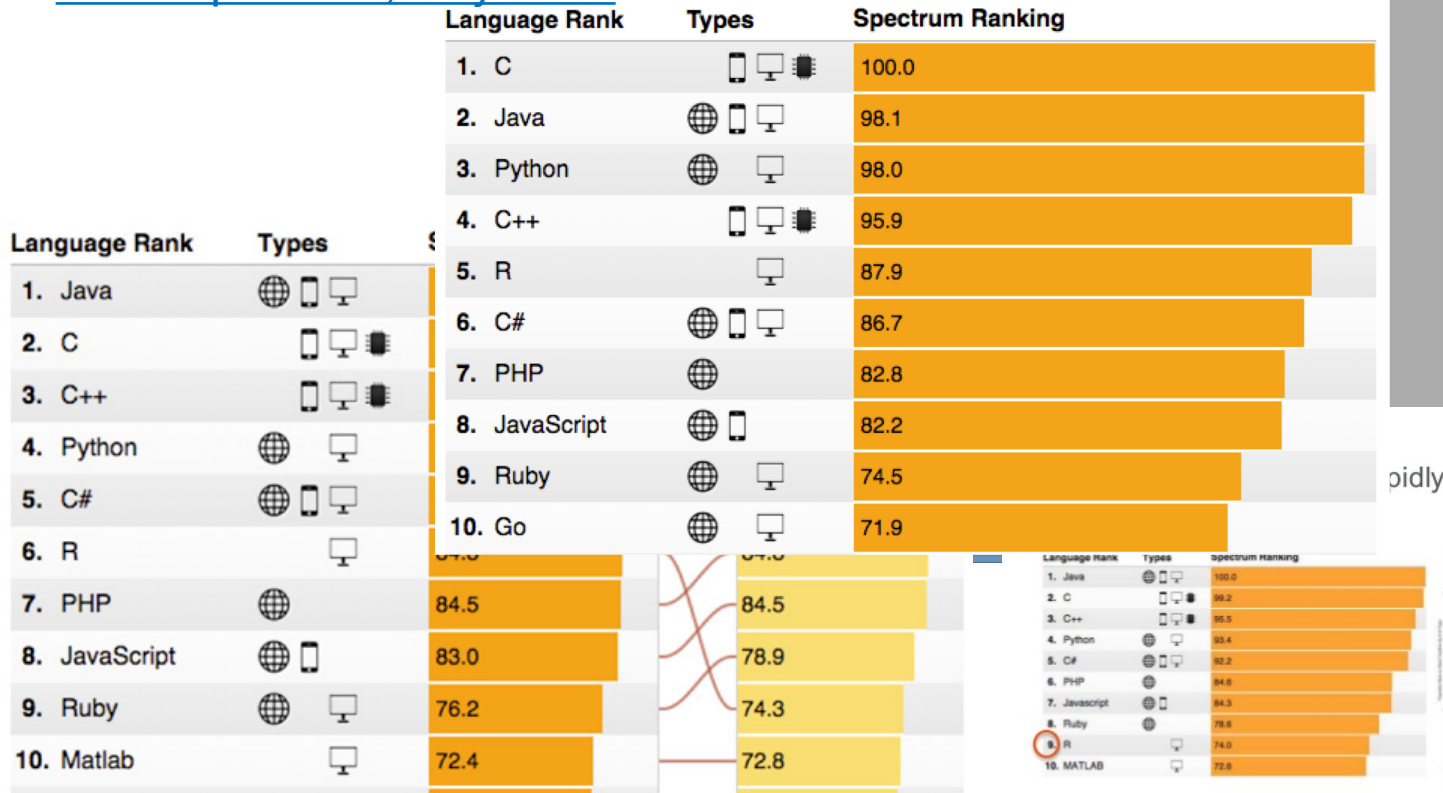
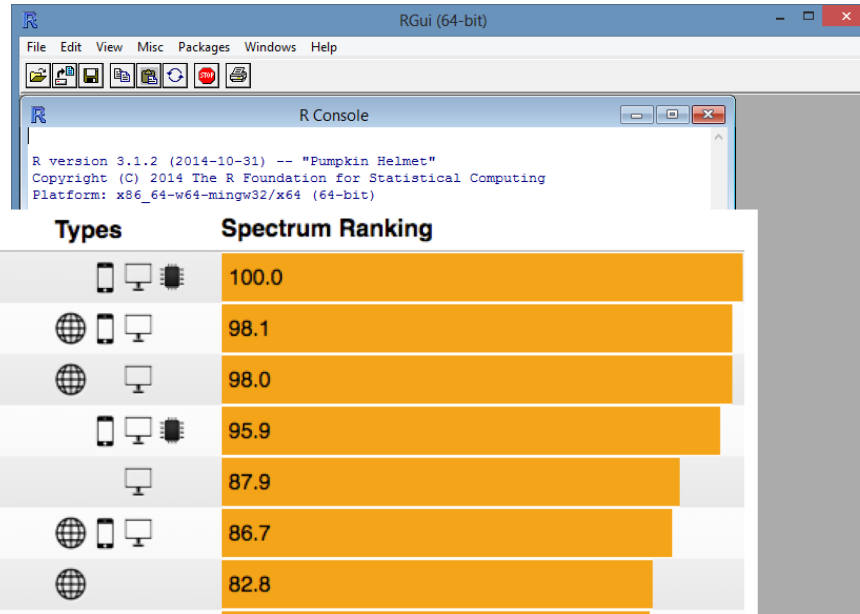
Marquette University logo and tagline: Be The Difference.

© 2013 Marquette University

R IS THE STATISTICAL SOFTWARE THAT WE MOSTLY USE IN THIS COURSE

- It is free!!

[IEEE Spectrum, July 2016](#)



#15: R



[IEEE Spectrum, July 2015](#)

- [IEEE Spectrum, July 2014](#)



- [RedMonk Programming Language Rankings, 2013](#)

HISTORY OF R (FROM [WIKIPEDIA](#))

- R is a [programming language](#) and software environment for [statistical computing](#) and graphics. The R language is widely used among [statisticians](#) and [data miners](#) for developing [statistical software](#) and data analysis. Polls, [surveys of data miners](#), and studies of scholarly literature databases show that R's popularity has increased substantially in recent years.
- R is an implementation of the [S programming language](#) combined with [lexical scoping](#) semantics inspired by [Scheme](#). S was created by [John Chambers](#) while at [Bell Labs](#). There are some important differences, but much of the code written for S runs unaltered.
- R was created by [Ross Ihaka](#) and [Robert Gentleman](#) at the [University of Auckland](#), New Zealand, and is currently developed by the *R Development Core Team*, of which Chambers is a member. R is named partly after the first names of the first two R authors and partly as a play on the name of [S](#).
- R is a [GNU project](#). The [source code](#) for the R software environment is written primarily in [C](#), [Fortran](#), and R. R is freely available under the [GNU General Public License](#), and pre-compiled binary versions are provided for various [operating systems](#). R uses a [command line interface](#); there are also several [graphical front-ends](#) for it.

R : THE STATISTICAL SOFTWARE

- Download and Install:

Windows

[Comprehensive R Archive Network](#)

Mac

The collage displays various R environments and data analysis results:

- R Console (Top Left):** Shows a payroll dataset with columns for employee ID, name, and salary. The data is displayed in a grid format.
- RStudio (Center):** The main interface showing a script editor with R code for loading the 'diamonds' dataset and creating a scatter plot titled 'Diamond Pricing'. The plot shows 'Price' on the y-axis (0 to 15,000) and 'Carat' on the x-axis (0.0 to 3.5). Points are colored by 'Clarity' (I1, SI2, SI1, VS2, VVS2, VVS1, IF).
- R Workspace Browser (Top Right):** A table listing objects in the workspace, including 'data', 'g', 'l', 'n', 'opar', 'pie.sales', 'pin', 'scale', 'usr', 'women', 'height', 'weight', and 'x'.
- R Package Manager (Bottom Right):** A window showing the status of installed packages, including 'graphics', 'grid', 'lattice', and 'methods'.

RStudio Script Editor Content:

```
1 library(ggplot2)
2 source("plots/formatPlot.R")
3
4 view(diamonds)
5 summary(diamonds)
6
7 summary(diamonds$price)
8 aveSize <- round(mean(diamonds$carat), 4)
9 clarity <- levels(diamonds$clarity)
10
11 p <- qplot(carat, price,
12            data=diamonds, color=clarity,
13            xlab="Carat", ylab="Price",
14            main="Diamond Pricing")
15
16 format.plot(p, size=24)
```

Console Output:

```
Min. x: 0.000 Min. y: 0.000 Min. z: 0.000
1st Qu.: 4.710 1st Qu.: 4.720 1st Qu.: 2.910
Median: 5.700 Median: 5.710 Median: 3.530
Mean: 5.731 Mean: 5.735 Mean: 3.539
3rd Qu.: 6.540 3rd Qu.: 6.540 3rd Qu.: 4.040
Max.: 10.740 Max.: 58.900 Max.: 31.800
```

Summary Statistics:

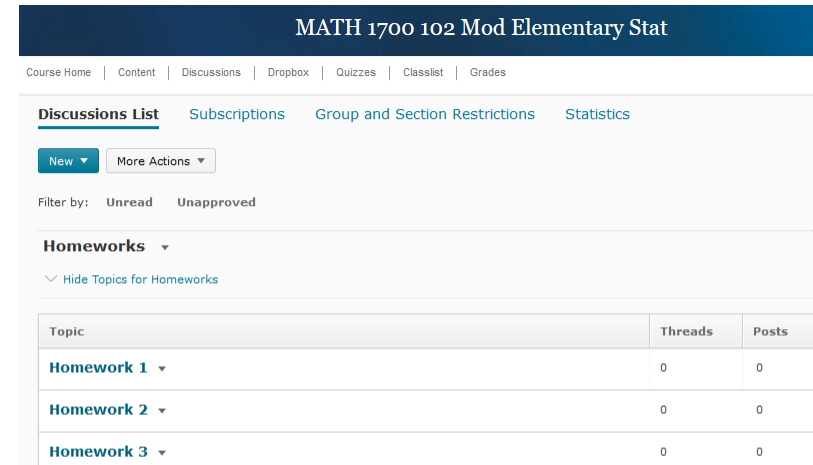
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
326	950	2401	3933	5324	18820

- RStudio:
 - [Download](#)

MATH 4780 CONT...

- Any General questions about Homework, **Projects** and Exams:

- **SHOULD** be posted in d21 Discussion Board.
- I will **NOT** answer general emails about Homework and Exams.



Topic	Threads	Posts
Homework 1	0	0
Homework 2	0	0
Homework 3	0	0

- Homework, Projects and Exams:**

- Should be submitted as a **PDF** file:
 - How to Combine Images into a PDF file [FREE & EASY + No Software] ([Youtube](#))
 - Microsoft Word to PDF in 10 Seconds ([Youtube](#))
 - How to: convert Images to PDF in Macbook/iMac ([Youtube](#))
 - <http://apple.stackexchange.com/questions/11163/how-do-i-combine-two-or-more-images-to-get-a-single-pdf-file>

1.1 REGRESSION AND MODEL BUILDING

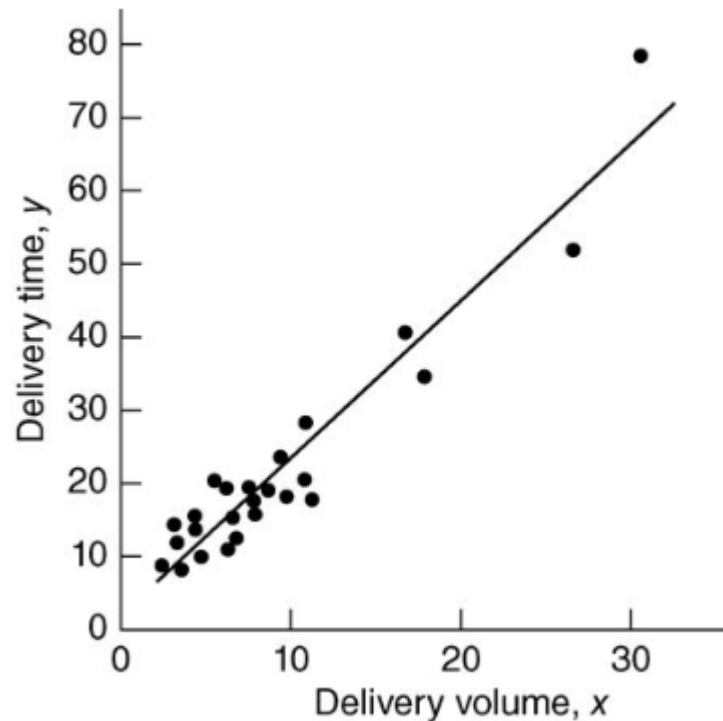
- A regression function describes how a response variable y changes as an explanatory variable x changes.
- We often use a **regression line** to predict the value of y for a given value of x .
- Example: How much should you pay for a house?
- What factors are important in determining a reasonable price?
 - Amenities
 - Location
 - **Square footage**
- To determine a price, you might consider a model of the form:

$$\text{Price} = f(\text{square footage}) + \epsilon$$

1.1 REGRESSION AND MODEL BUILDING

- Delivery time example: We want to study the relationship between the delivery time and delivery volume.
- If we let y represent delivery time and x represent delivery volume, then the equation of a straight line relating these two variables is

$$y = \beta_0 + \beta_1 x + \varepsilon$$



1.1 REGRESSION AND MODEL BUILDING

- Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where

- y : dependent (response) variable
- x : independent (regressor/predictor) variable
- β_0 : intercept
- β_1 : slope
- ε : random error term

R CODE

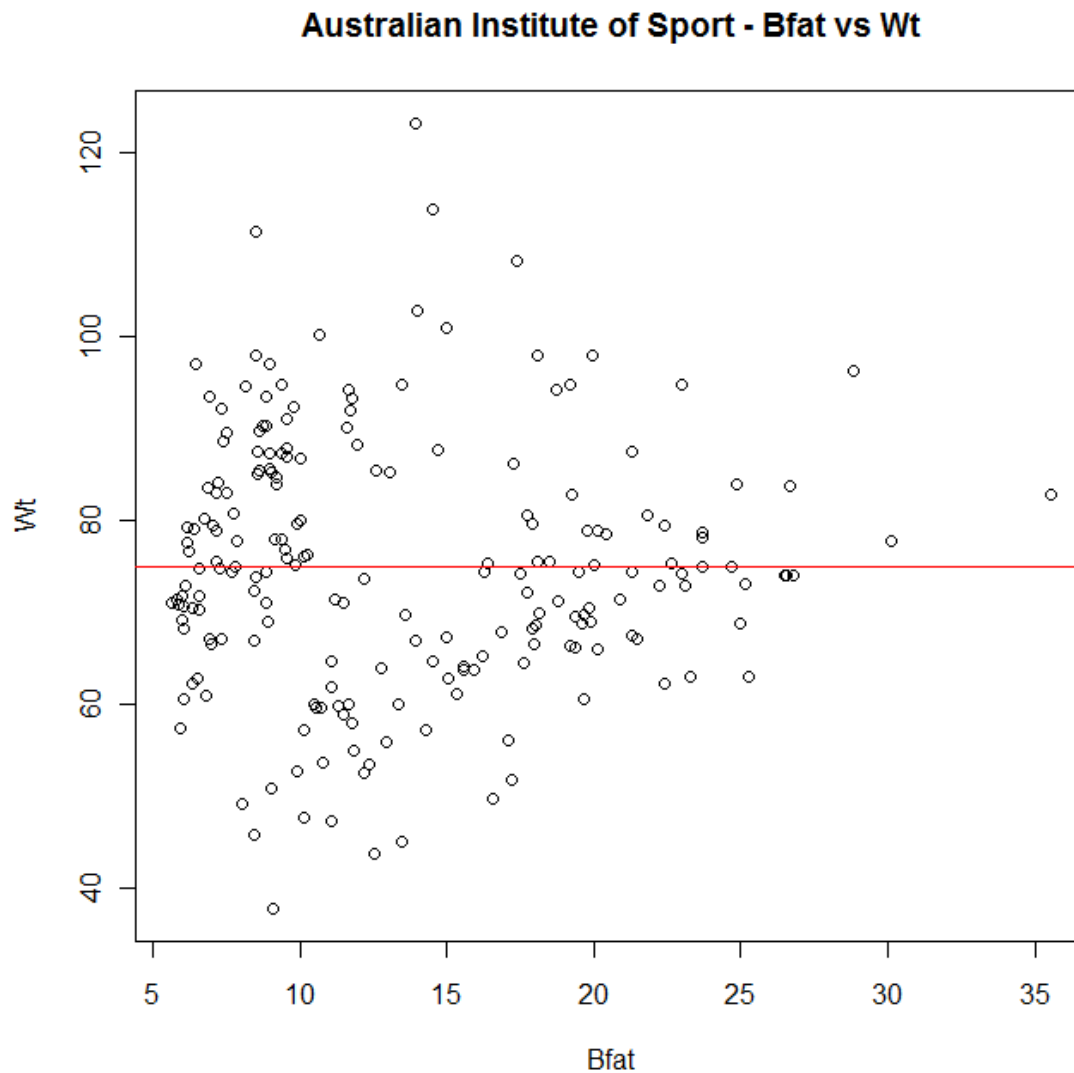
- `delivery=read.table("data-ex-2-9_(Delivery_Time).txt", header=T)`
- `y=delivery$Delivery.Time..y`
- `x=delivery$Number.of.Cases..x`
- `plot(x, y, xlab="Delivery volume, x", ylab="Delivery time, y")`
- `f=lm(y~x)`
- `abline(f)`

EXAMPLE – AUSTRALIAN INSTITUTE OF SPORT

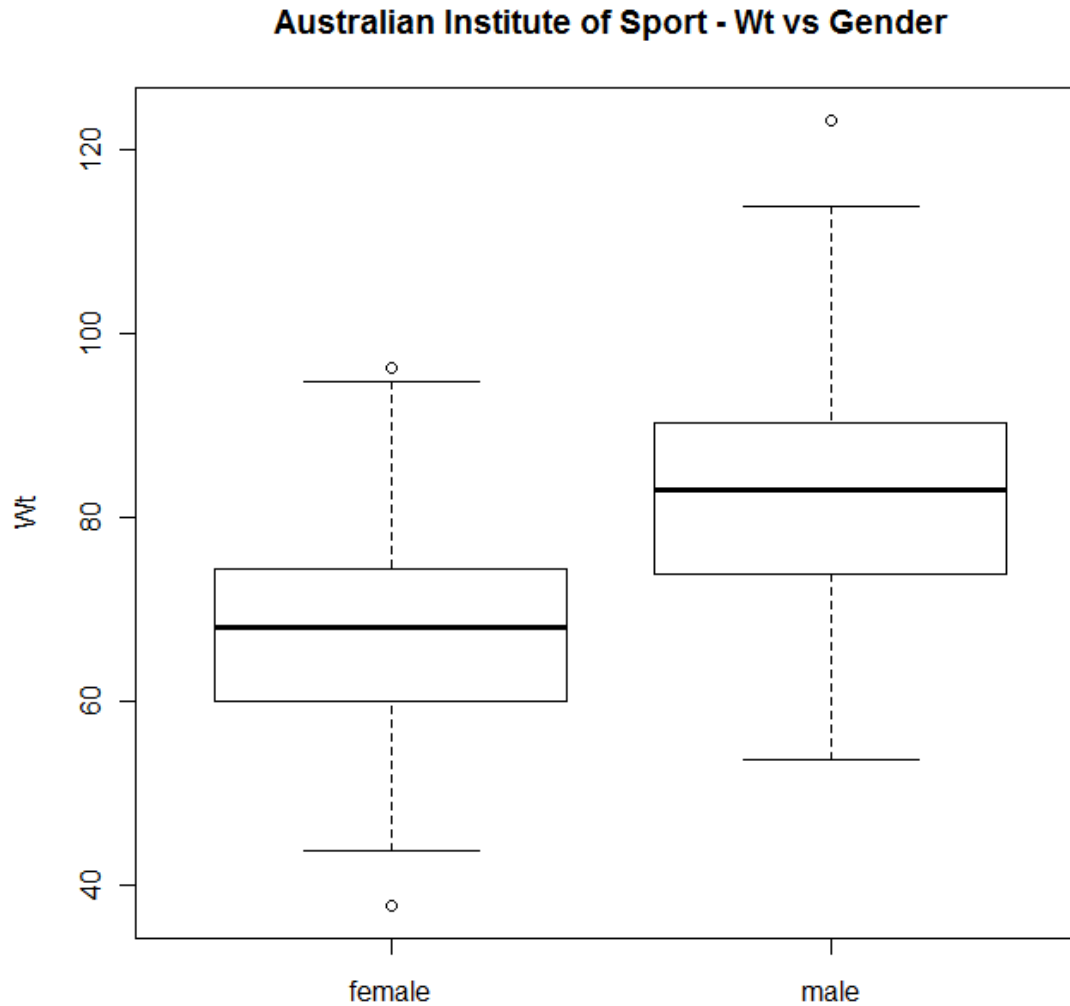
- Data on 102 male and 100 female athletes collected at the Australian Institute of Sport, (courtesy of Richard Telford and Ross Cunningham.)

	Gender	Bfat	Wt
• 1	female	19.75	78.9
• 2	female	21.30	74.4
• 3	female	19.88	69.1
• 4	female	23.66	74.9
• 5	female	17.64	64.6
	:	:	:
• 198	male	11.79	93.2
• 199	male	10.05	80.0
• 200	male	8.51	73.8
• 201	male	11.50	71.1
• 202	male	6.26	76.7

EXAMPLE CONT'D

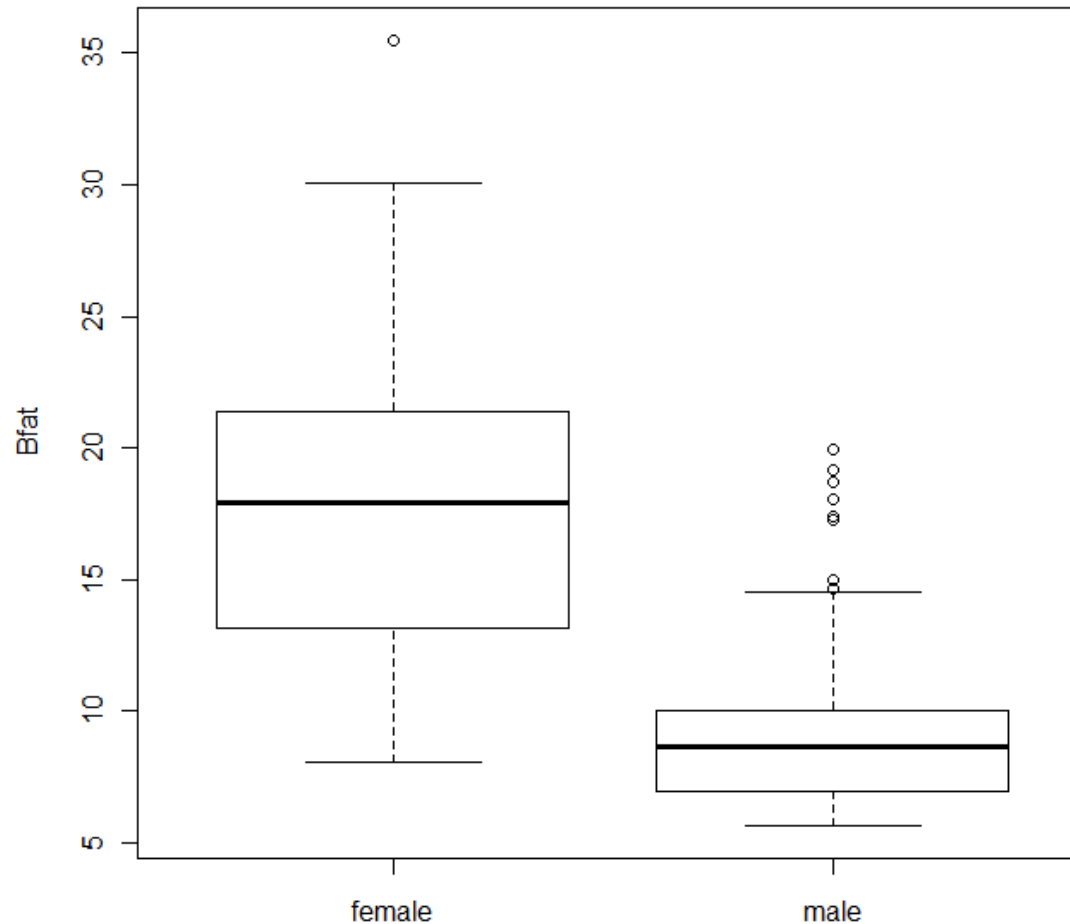


EXAMPLE CONT'D

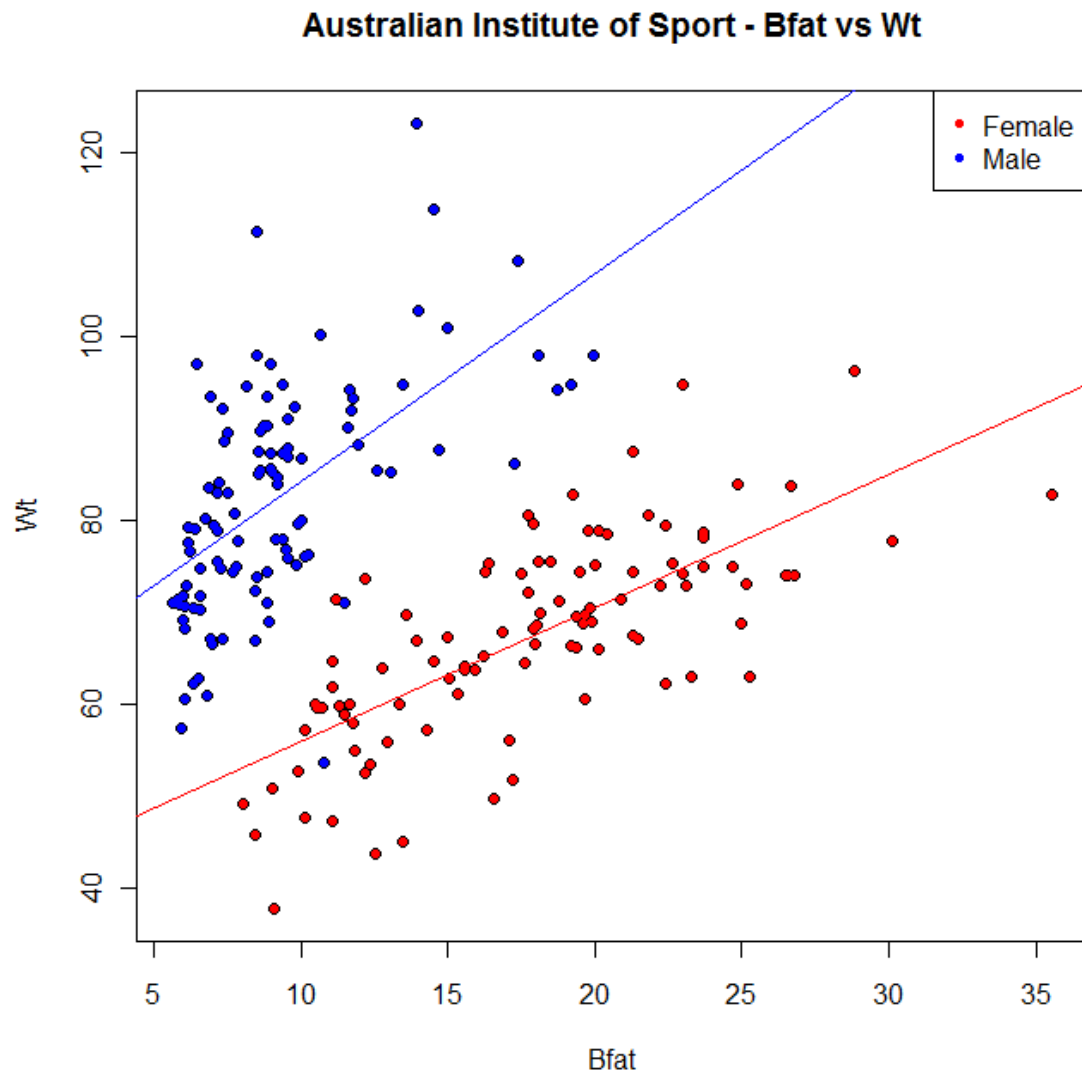


EXAMPLE CONT'D

Australian Institute of Sport - Bfat vs Gender



EXAMPLE CONT'D



R CODE

- `load("ais.rda")`
- `plot(ais$Bfat, ais$Ht, xlab="Bfat", ylab="Wt")`
- `abline(lm(ais$Wt~ais$Bfat),col=2)`
- `boxplot(ais$Wt~ais$sex)`
- `boxplot(ais$Bfat~ais$sex)`
- `plot(ais$Bfat, ais$Wt, xlab="Bfat", ylab="Wt")`
- `points(ais$Bfat[ais$sex!="male"],ais$Wt[ais$sex!="male"],pch=20,col=2)`
- `points(ais$Bfat[ais$sex=="male"],ais$Wt[ais$sex=="male"],pch=20,col=4)`
- `abline(lm(ais$Wt[ais$sex!="male"]~ais$Bfat[ais$sex!="male"]),col=2)`
- `abline(lm(ais$Wt[ais$sex=="male"]~ais$Bfat[ais$sex=="male"]),col=4)`

ASSESSING THE ABILITY OF NFL KICKERS

REF: S.J. SHEATHER, *A MODERN APPROACH TO REGRESSION WITH R*,

- In the Keeping Score column by Aaron Schatz in the Sunday November 12, 2006 edition of the *New York Times* entitled “**N.F.L. Kickers Are Judged on the Wrong Criteria**” the author makes the following claim:
- There is effectively no correlation between a kicker’s field goal percentage one season and his field goal percentage the next.
- We can show that once the different ability of field goal kickers is taken into account, there is a highly statistically significant **negative correlation** between a kicker’s field goal percentage one season and his field goal percentage the next.

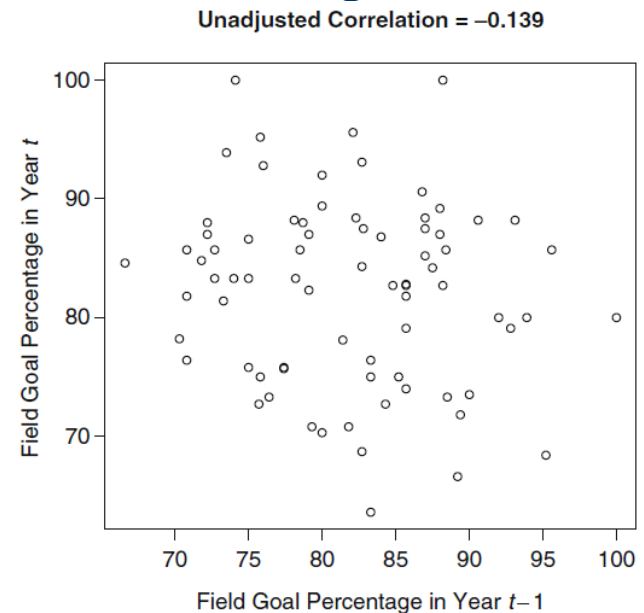


Figure 1.1 A plot of field goal percentages in the current and previous year

ASSESSING THE ABILITY OF NFL KICKERS (CONT...)

REF: S.J. SHEATHER, *A MODERN APPROACH TO REGRESSION WITH R*,

- In order to examine the claim we consider data on the 19 NFL field goal kickers who made at least ten field goal attempts in each of the 2002, 2003, 2004, 2005 and 2006 seasons.
- It can be shown that the resulting correlation in Figure 1.1 of -0.139 is not statistically significantly different from zero (p -value = 0.230). This result is in line with Schatz's claim of "effectively no correlation." However, this approach is **fundamentally flawed** as it fails to take into account the potentially *different abilities of the 19 kickers*.
- There is very strong evidence that the intercepts of the 19 lines differ (p -value = 0.006) but little evidence that the slopes of the 19 lines differ (p -value = 0.939).
- Details on how to perform these calculations will be provided in Ch. 3.

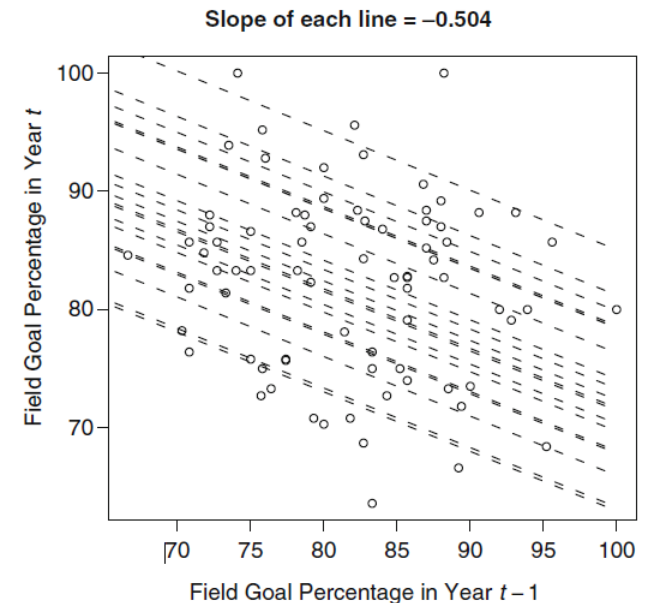


Figure 1.2 Allowing for different abilities across the 19 field goal kickers

1.1 REGRESSION AND MODEL BUILDING

- **Regression analysis** is a statistical technique for investigating and modeling the relationship between variables.
- Equation of a straight line (classical)

$$y = mx + b$$

we usually write this as

$$y = \beta_0 + \beta_1 x$$

- Not all observations will fall exactly on a straight line.

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where ε represents error

- it is a random variable that accounts for the failure of the model to fit the data *exactly*.
- $\varepsilon \sim N(0, \sigma^2)$

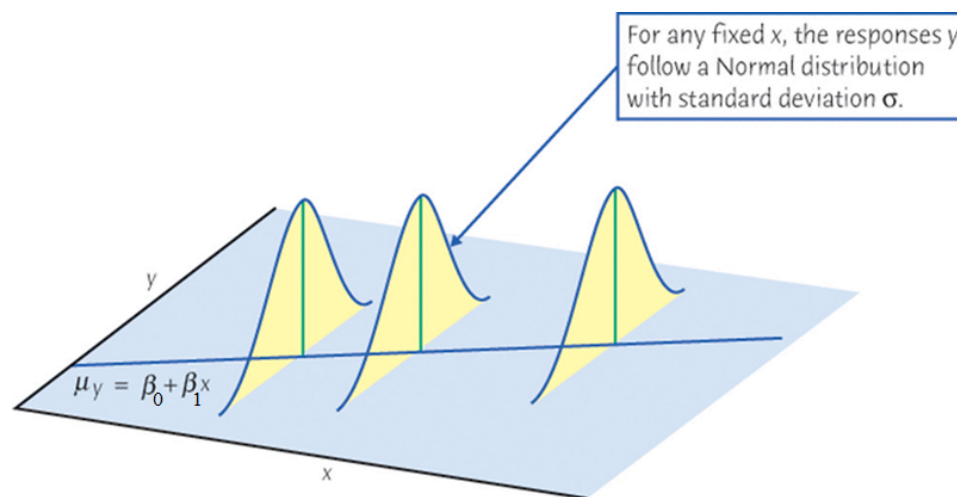
1.1 REGRESSION AND MODEL BUILDING

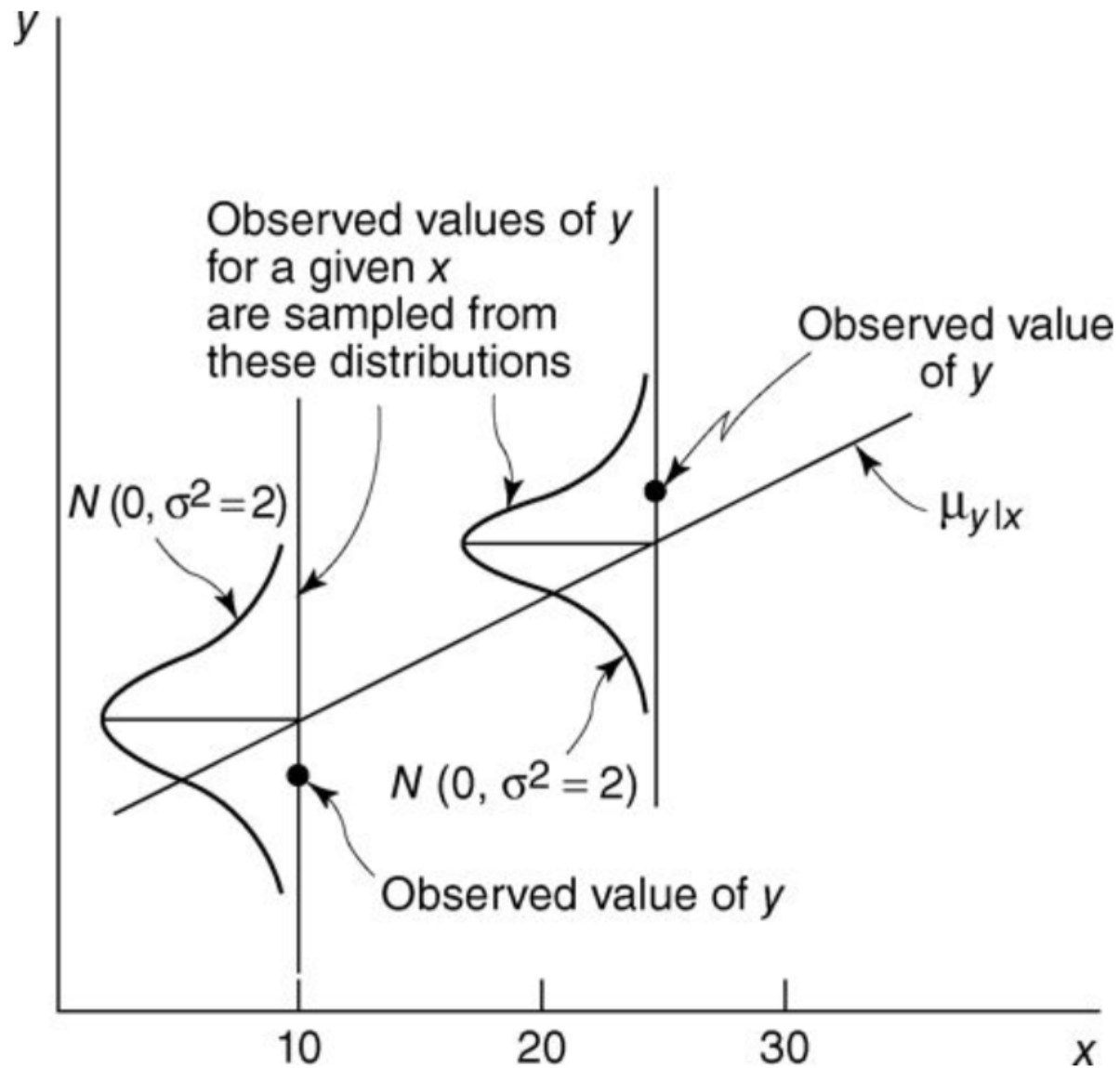
- The **mean response** at any value, x , of the regressor variable is

$$E(y|x) = \mu_{y|x} = E(\beta_0 + \beta_1 x + \varepsilon) = \beta_0 + \beta_1 x$$

- The **variance** of y at any given x is

$$\text{Var}(y|x) = \sigma_{y|x}^2 = \text{Var}(\beta_0 + \beta_1 x + \varepsilon) = \sigma^2$$





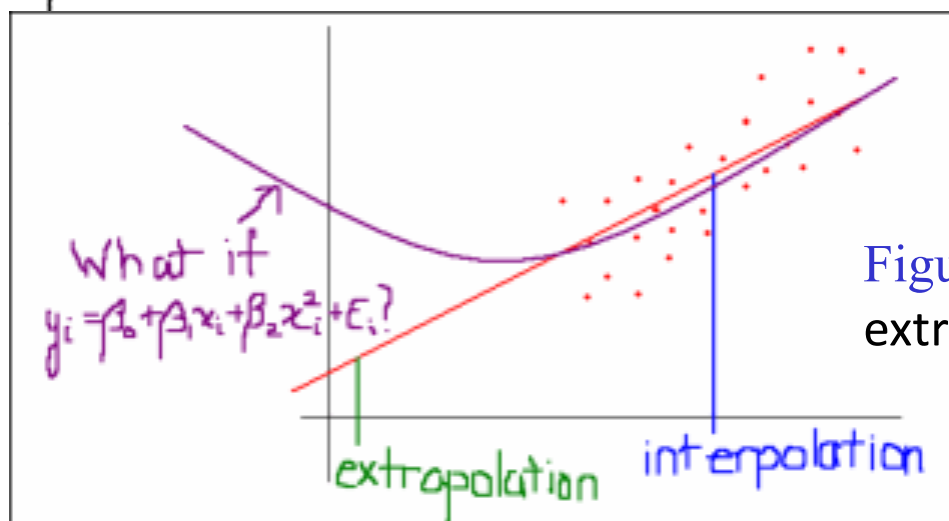
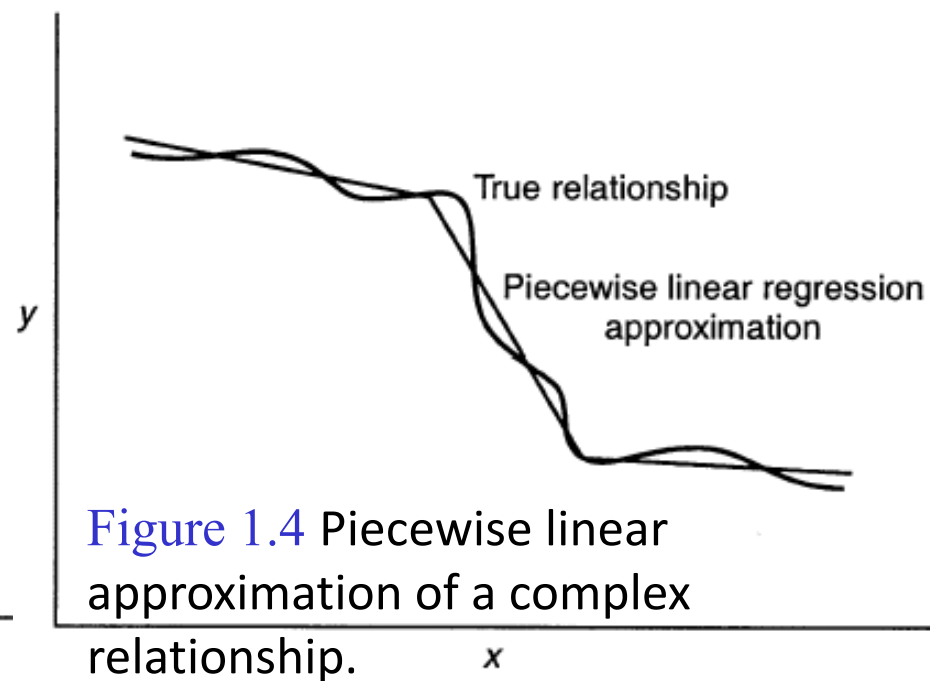
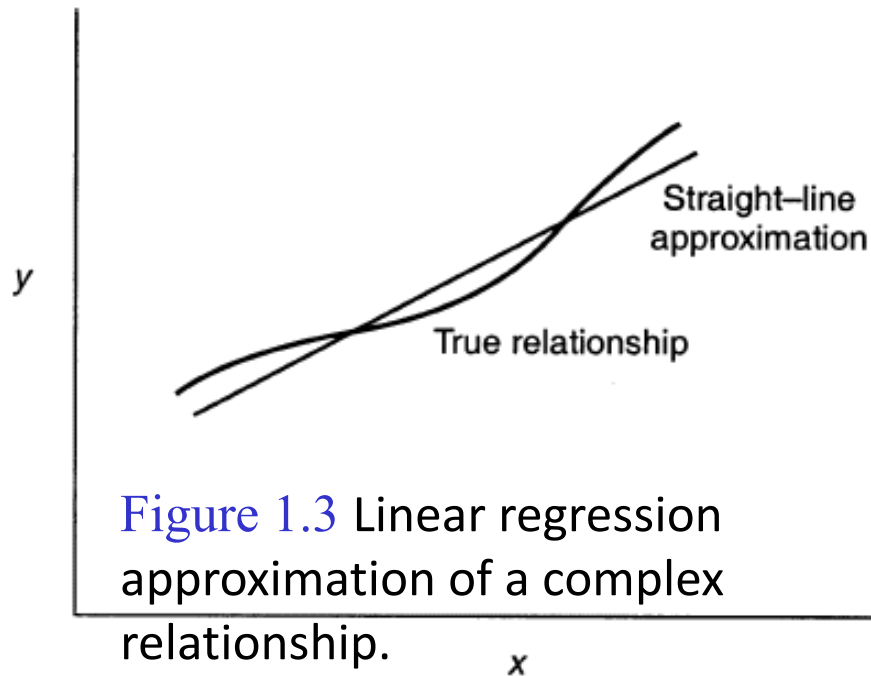


Figure 1.5 The danger of extrapolation in regression.

MENU PRICING IN A NEW RESTAURANT IN NY CITY

REF: S.J. SHEATHER, A MODERN APPROACH TO REGRESSION WITH R,

- You have been asked to determine the pricing of the restaurants dinner menu such that it is competitively positioned with other high-end Italian restaurants in the target area

Y = Price = the price (in \$US) of dinner (including one drink & a tip)

x_1 = Food = customer rating of the food (out of 30)

x_2 = Décor = customer rating of the decor (out of 30)

x_3 = Service = customer rating of the service (out of 30)

x_4 = East = dummy variable = 1 (0) if the restaurant is east (west) of Fifth Avenue

- In particular one may ask you to
 - Develop a regression model that *directly predicts* the price of dinner (in dollars) using a subset or all of the four potential predictor variables listed above.
 - Determine which of the predictor variables Food, Décor and Service has the largest estimated effect on Price? Is this effect also the most statistically significant?
 - If the aim is to choose the location of the restaurant so that the price achieved for dinner is maximized, should the new restaurant be on the east or west of Fifth Avenue?
 - Does it seem possible to achieve a price premium for “setting a new standard for high-quality service in Manhattan” for Italian restaurants?
 - Identify the restaurants in the data set which, given the customer ratings, are (i) unusually highly priced; and (ii) unusually lowly priced.

MENU PRICING IN A NEW RESTAURANT IN NY CITY

REF: S.J. SHEATHER, *A MODERN APPROACH TO REGRESSION WITH R*,

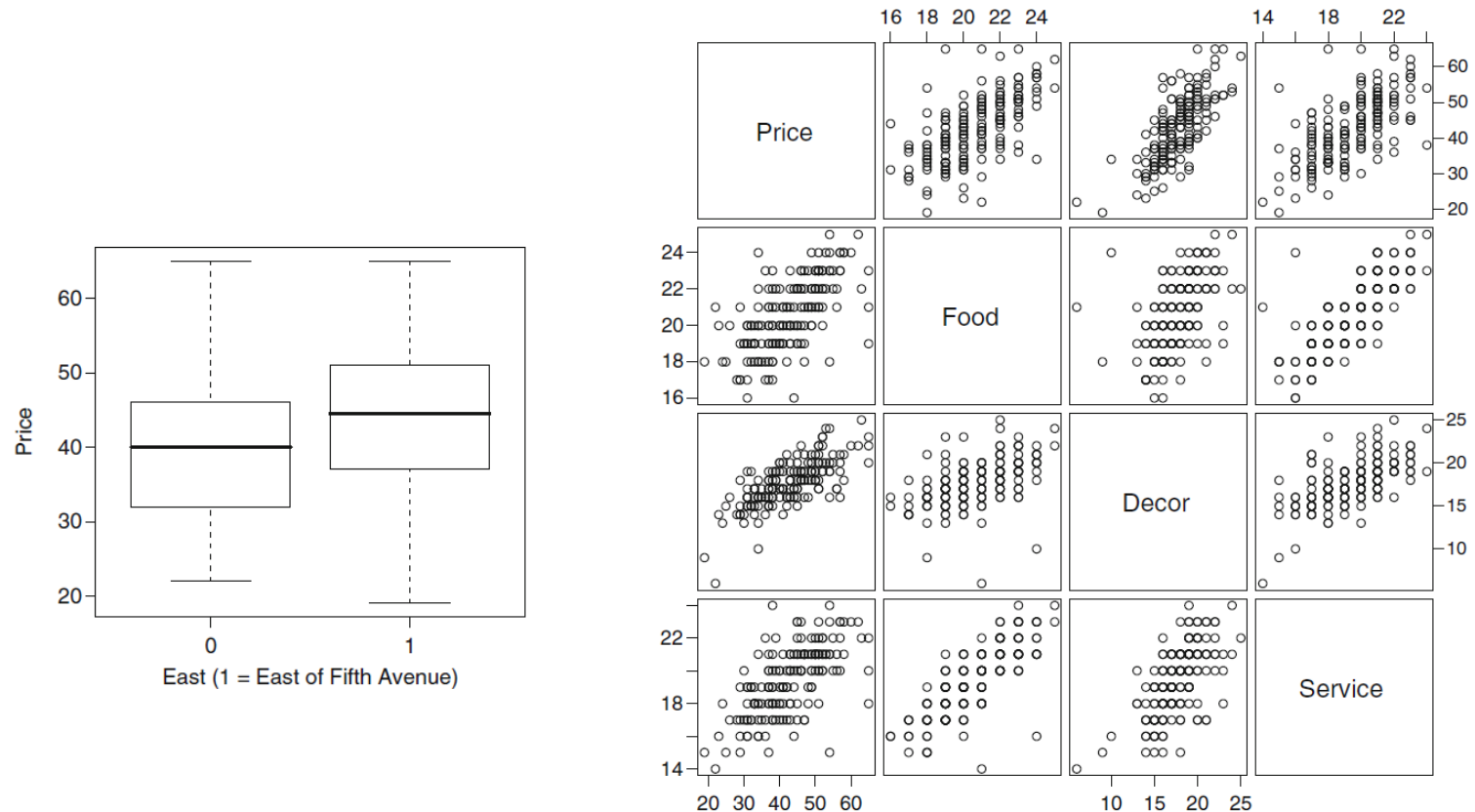


Figure 1.5 Matrix plot of Price, Food, Décor and Service ratings

- Multiple Linear Regression Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

1.2 DATA COLLECTION

- Three different methods can be used for data collection
 1. A retrospective study based on historical data
 - Pitfalls
 - Some of the relevant data often are missing.
 - The reliability and quality of the data are often highly questionable.
 - The nature of the data often may not allow us to address the problem at hand.
 - The analyst often tries to use the data in ways they were never intended to be used.
 - Logs, notebooks, and memories may not explain interesting phenomena identified by the data analysis.
 2. An observational study
 - Pitfall: Spurious Correlation Due to Omitted Variables
 3. A designed experiment

1.3 USES OF REGRESSION

There are many uses of regression, including:

1. Data description
2. Parameter estimation
3. Prediction and estimation
4. Control

Regression analysis is perhaps the most widely used statistical technique, and probably the most widely misused.

1.3 USES OF REGRESSION

Cause and Effect Relationships

- Caution: just because you *can* fit a linear model to a set of data, does not mean you should.
- It is relatively easy to build “nonsense” relationships between variables
- A perceived relationship between a **response** variable and an **explanatory** variable that has been misestimated due to the failure to account for a confounding factor (lurking variable) is termed a spurious relationship
- Examples of spurious relationship
- Regression does not necessarily imply causality

LURKING VARIABLE AND SIMPSON'S PARADOX

- **Lurking variable:** A variable that is not included in a study but has an effect on the variables of the study and makes it appear that those variables are related.
- **Simpson's Paradox:** An association or comparison that holds for all of several groups can **reverse direction** when a **lurking variable** is present.
- **Example: Kidney stone treatment**(*Br Med J (Clin Res Ed)* 292 (6524): 879–882)

	Treatment A	Treatment B
Small Stones	<i>Group 1</i> 93% (81/87)	<i>Group 2</i> 87% (234/270)
Large Stones	<i>Group 3</i> 73% (192/263)	<i>Group 4</i> 69% (55/80)
Both	78% (273/350)	83% (289/350)

- http://en.wikipedia.org/wiki/Simpson's_Paradox

MODEL BUILDING IN REGRESSION

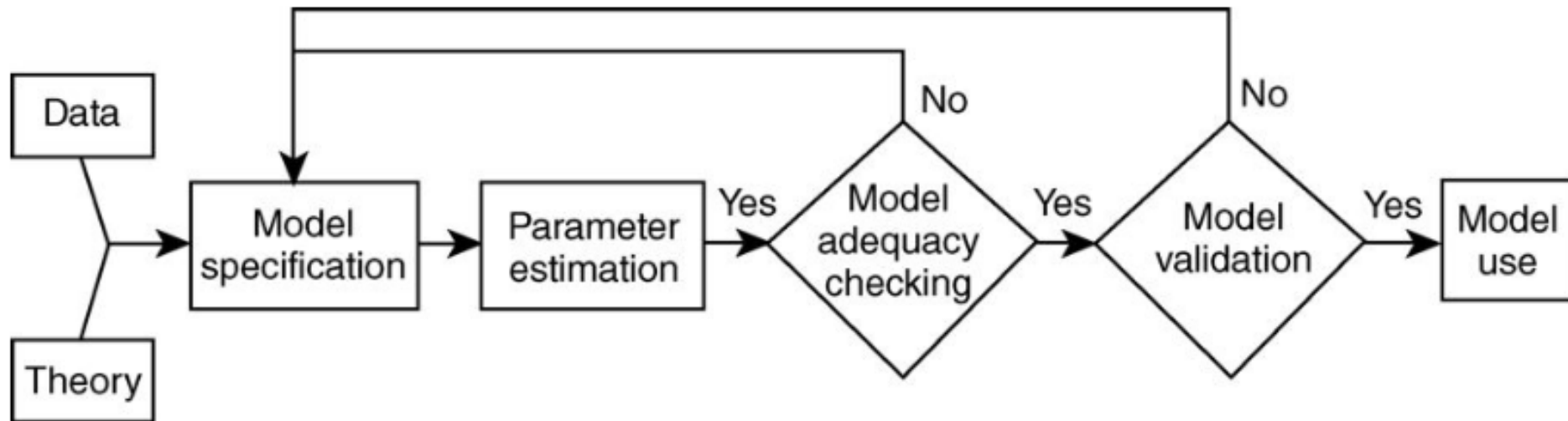


Figure 1.8 Regression model-building process.

QUESTIONS?

- ANY QUESTION?